

1. Administrative & Study Identification

Full Study Title: A Randomised, Double-blind, Placebo-controlled, Parallel Group, Multicentre, Phase III Study to Evaluate the Efficacy and Safety of Tezepelumab in Adult Participants With Moderate to Very Severe Chronic Obstructive Pulmonary Disease
Short Title/Acronym: A Study to Investigate the Efficacy and Safety of Tezepelumab in Adult Participants With Moderate to Very Severe COPD
Protocol Number: D5241C00007 **NCT Number:** NCT06878261
Sponsor/Company Name: AstraZeneca **Investigational Product:** Tezepelumab (Drug Preferred Name: tezepelumab) **Trade Name:** Tezspire **ATC Code:** R03DX11 **Phase of Development:** Phase III **Trial Status:** RECRUITING (Currently recruiting participants) **Medical Monitor:** [AstraZeneca Clinical Medical Monitor / Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the efficacy of tezepelumab compared to placebo in reducing the annualised rate of moderate or severe COPD exacerbations up to the 76-week treatment period. **Secondary Objectives:** To evaluate the effect of tezepelumab on lung function (change from baseline in post-BD and pre-BD FEV1) and patient-reported outcomes/quality of life (change from baseline in SGRQ total score).

2.2 Background & Rationale To address the unmet medical need in patients with moderate to very severe Chronic Obstructive Pulmonary Disease (COPD) who continue to experience frequent exacerbations (≥ 2 moderate or ≥ 1 severe in the prior year) despite receiving standard-of-care triple (ICS+LABA+LAMA) or dual inhaled maintenance therapy. The study investigates whether the biologic tezepelumab can safely and effectively reduce exacerbations and improve respiratory function in this high-risk population with eosinophilic inflammation (≥ 150 cells/ μ L).

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Parallel Group (tezepelumab dose 1 vs. tezepelumab dose 2 vs. placebo) **Criteria Type:** PHASE_REQUIREMENT (Trial must be in PHASE3)

3.2 Planned Enrollment & Duration **Targeted Enrollment:** 990 evaluable patients **Number of Sites:** Multicenter (Global)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female, ≥ 40 to ≤ 80 years old. 2. COPD diagnosis ≥ 1 year. 3. Post-bronchodilator (BD) FEV1 $\geq 20\%$ and $\leq 70\%$ predicted normal, FEV1/FVC < 0.70 at screening. 4. On Triple (ICS+LABA+LAMA) or dual inhaled COPD therapy for ≥ 3 consecutive months prior to Visit 1. 5. History of ≥ 2 moderate or ≥ 1 severe COPD exacerbations in the prior year (at least 1 of 2 moderate must require systemic corticosteroids, and at least one must have occurred while on triple/dual therapy). 6. Blood eosinophil (EOS) count ≥ 150 cells/ μ L during screening. 7. COPD Assessment Test (CAT) score ≥ 15 at screening. 8. Former or current smokers with ≥ 10 pack-years history.

4.2 Exclusion Criteria 1. Clinically important pulmonary disease or radiological findings suggestive of a respiratory disease other than COPD. 2. Asthma, including pediatric asthma, or Asthma-COPD Overlap Syndrome (ACOS). 3. Any unstable disorder that can impact participant safety or study outcomes. 4. Tuberculosis requiring treatment within 12 months prior to Visit 2. 5. Current or past malignancies. 6. Concomitant use of macrolides (< 6 months) or systemic immuno-suppressive/immuno-modulating medications. 7. Long-term oxygen therapy (LTOT) > 4.0 L/min or SO_2 saturation $< 89\%$ despite LTOT.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Monthly injection of one of two different doses of tezepelumab, or matching placebo. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Minimum of 52 weeks up to a maximum of 76 weeks of exposure, followed by a 12-week off-treatment safety follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Maintenance triple (ICS+LABA+LAMA) or dual inhaled COPD therapy. **Prohibited:** Initiation of macrolides, and systemic immuno-suppressive or immuno-modulating medications.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Spirometry (FEV1), Eosinophil Labs, CAT Questionnaire
Baseline	Day 0	Randomization, First SC Dose, Baseline SGRQ	Interim Monthly Investigational

Product Administration, Safety Labs, Efficacy Assessments (Spirometry/Exacerbation Tracking) | | **End of Study** | Week 88 | End of 76-week treatment + 12-week safety follow-up, Final Vital Signs, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualised rate of moderate or severe COPD exacerbations up to the 76-week treatment period compared to placebo. **Measurement Method:** Based on moderate/severe exacerbations as reported by the investigator.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in post-BD FEV1. 2. Change from baseline in pre-dose/pre-bronchodilator (pre-BD) FEV1. 3. Change from baseline in the St. George's Respiratory Questionnaire (SGRQ) total score.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient reporting and investigator assessment during monthly onsite visits and the 12-week off-treatment safety follow-up. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor/regulatory authorities. **Data Monitoring Committee (DMC):** External independent DMC established to review unblinded safety and efficacy data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy endpoint analysis; Safety Set for all safety/AE summaries. **Sample Size Justification:** $N = 990$, adequately powered to detect a statistically significant reduction in the annualized exacerbation rate between tezepelumab and placebo. **Handling of Missing Data:** Handled via multiple imputation models and mixed-effects models for repeated measures (MMRM) for continuous secondary endpoints (FEV1, SGRQ).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence, accurate AE reporting, and IP accountability. **Audit Trail:** Strict eCRF tracking, secure randomization via IRT (Interactive Response Technology), and standard data clarification workflows.

11. Study Identifiers & Governance

Posting ID: 205495743248177483 **Registry Number:** NCT06878261

Sponsor/Lead Organization: AstraZeneca **Study Title:** A Study to Investigate the Efficacy and Safety of Tezepelumab in Adult Participants With Moderate to Very Severe COPD (D5241C00007)

Official Regulatory Title: A Randomised, Double-blind, Placebo-controlled, Parallel Group, Multicentre, Phase III Study to Evaluate the Efficacy and Safety of Tezepelumab in Adult Participants With Moderate to Very Severe Chronic Obstructive Pulmonary Disease **Source Level:** 5 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/tezspire-epar-product-information_en.pdf

12. Clinical Framework (COPD Specific)

Disease Areas: Chronic Obstructive Pulmonary Disease (COPD) **Name:** COPD (Chronic Obstructive Pulmonary Disease) **Preferred Name:**

Chronic Obstructive Airway Disease **Preferred UMLS Name:** Chronic Obstructive Pulmonary Disease **Semantic Type:** Disease or Syndrome

Definition: A disease of chronic diffuse irreversible airflow obstruction. Subcategories of COPD include CHRONIC BRONCHITIS and PULMONARY EMPHYSEMA. **Diagnosis Threshold:** Post-BD FEV1 $\geq 20\%$ and $\leq 70\%$ PN, FEV1/FVC < 0.70 ; EOS ≥ 150 cells/ μ L

Medical Terminologies: MeSH Code: D029424 | HPO Code: HP:0006510 | SNOMED ID: 13645005 **SNOMED Hierarchy:** 17097001|Chronic disease

of respiratory system|Chronic disease of respiratory system (disorder), 27624003|Chronic disease|Chronic disease (disorder), 50043002|Disorder of respiratory system|Disorder of respiratory system (disorder), 362965005|Disorder of body system|Disorder of body system (disorder), 64572001|Disease|Disease (disorder), 233776003|Tracheobronchial disorder|Tracheobronchial disorder (disorder), 128272009|Disorder of lower respiratory system|Disorder of lower respiratory system (disorder), 90734009|Chronic|Chronic (qualifier value), 91724006|Tracheobronchial structure|Tracheobronchial structure (body structure), 82094008|Lower respiratory tract structure|Lower respiratory tract structure (body structure), 400141005|Lower respiratory system structure|Lower respiratory system structure (body structure), 20139000|Structure of respiratory system|Structure of respiratory system (body structure), 91689009|Body system structure|Body system structure (body structure), 321667001|Respiratory tract structure|Respiratory tract structure (body structure), 714323000|Structure of organ in respiratory system|Structure of organ in respiratory system (body structure), 113343008|Body organ structure|Body organ structure (body structure), 118969007|Respiratory system subdivision|Structure of respiratory system subdivision (body structure), 91690000|Entire subdivision of organ system|Entire subdivision of organ system (body structure), 708528005|Narrowing|Narrowing (morphologic abnormality) **Medical Methodology:** Biologic therapeutic intervention added to

background maintenance therapy **Optimization Target:** Reduction of annualised moderate-to-severe exacerbation rate

13. Study Procedures & Methodology

Management Pathway: Standard-of-care inhaled maintenance (Triple or Dual therapy) + SC Biologic **Education Protocol (HCP):** Protocol specific training, exacerbation criteria definitions, and SC administration training **Education Protocol (Patient):** Inhaler technique verification, exacerbation symptom recognition, and reporting instructions **Quality Audit Mechanism:** Centralized spirometry over-read and investigator-reported exacerbation adjudication

14. Observation & Follow-Up Schedule

Total Duration: Up to 88 weeks (Maximum 76 weeks treatment + 12 weeks follow-up) **Onsite Visit Cadence:** Every 4 weeks (monthly) for SC dosing and clinical evaluation **Remote Monitoring Frequency:** Intermittent phone contacts as dictated by the clinical monitoring plan **Checklist Review Interval:** SGRQ and CAT assessments performed at scheduled benchmark visits

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				